

ML Notes2

1. Bias and Variance:

When building a machine learning model, we want it to learn the actual pattern in the data without memorizing noise.

Two major sources of prediction error are:

- Bias
- Variance

Bias:

Bias is the error caused by making overly simple assumptions about the data.

A high-bias model fails to capture the underlying relationship.

Characteristics:

- Model is too simple
- Underfits the data
- Performs poorly on both training and testing data

Variance:

Variance is the amount by which the model changes when trained on different datasets.

A high-variance model learns not only the pattern but also the noise.

Characteristics

- Model is too complex
- Overfits the data
- Excellent training accuracy
- Poor testing accuracy

Bias–Variance Tradeoff:

There is always a balance between bias and variance.

As model complexity increases:

- Bias decreases
- Variance increases

As model complexity decreases:

- Bias increases
- Variance decreases

Goal:

Find the optimal complexity where total prediction error is minimum.

Total Error:

Prediction Error

= Bias(squared) + Variance + Irreducible Error

where:

- Bias(squared) → error due to wrong assumptions
- Variance → error due to sensitivity to data
- Irreducible Error → noise that cannot be removed

What is Regularization?

Regularization is a technique used to reduce overfitting by adding a penalty to the model's coefficients. Instead of minimizing only the prediction error, we also penalize large coefficients. Regularization helps the model become simpler and generalize better.

Benefits:

- Prevents overfitting
- Improves generalization
- Reduces model complexity
- Handles multicollinearity
- Produces more stable predictions

Ridge Regression:

Ridge Regression adds the square of coefficients as a penalty. Large coefficients are reduced. No coefficient becomes exactly zero.

Advantages

- Reduces overfitting
- Works well with multicollinearity
- Uses all features
- Stable solution

Disadvantages

- Does not perform feature selection
- Less interpretable than LASSO

LASSO Regression:

Least Absolute Shrinkage and Selection Operator. Some coefficients become exactly zero.

Advantages

- Reduces overfitting
- Performs feature selection
- Produces simpler models
- Easy to interpret

Disadvantages

- Can remove useful correlated features
- May be unstable when features are highly correlated

Elastic Net Regression:

Elastic Net combines both **Ridge** and **LASSO** regularization.

Why Elastic Net?

Suppose a dataset has:

- Thousands of features
- Many correlated variables

LASSO may remove useful variables. Ridge keeps all variables. Elastic Net balances both approaches.

Advantages

- Performs feature selection
- Handles multicollinearity
- More stable than LASSO
- Suitable for high-dimensional datasets

Disadvantages

- Requires tuning two parameters
- Slightly more computationally expensive